

Topic: Amazon S3 (Simple Storage Service)

Easy English + Tenglish Revision Notes | 21 September 2026

1. What is S3?

English: Amazon S3 (Simple Storage Service) is an AWS **object storage** service used to store and retrieve data at scale.

Tenglish: S3 ante AWS lo files/data ni **objects** ga store cheyadaniki use chese service. Idi normal disk kaadu; object storage.

- Highly durable — **11 nines (99.999999999%)** durability.
- Practically unlimited scalability.
- Accessible over HTTP/HTTPS.
- No server management required.

Memory: S3 = Store files/data in AWS.

2. Basic Terminology

Term	Easy English	Tenglish
Bucket	Container that stores objects.	Files/objects ni store chese container.
Object	A file stored in S3, with data + metadata + key.	S3 lo store chesina file.
Key	Unique name/path of the object.	Object/file yokka unique name or path.
Region	AWS Region where the bucket is created.	Bucket create chesina AWS Region.

3. S3 Versioning

English: Versioning keeps multiple versions of the same object. It protects against accidental delete and accidental overwrite.

Tenglish: Same file ni multiple times update/upload chesina, old versions ni S3 preserve chestundi.

Every new upload can create a new Version ID. States in the source notes: **Enabled, Suspended, Never enabled.**

Memory: Versioning = Old versions safe.

4. Delete Marker

English: When versioning is enabled, deleting an object creates a **Delete Marker**. The object appears deleted, but older versions still exist.

Tenglish: Versioning ON unte delete chesina ventane old data destroy avvadu. Delete Marker create avtundi; old versions underneath remain.

Memory: "Delete" just hides the object; it does not destroy the old versions.

5. S3 Lifecycle Policy

English: Lifecycle policies automatically move or delete objects based on time-based rules. They are mainly used for cost optimization.

Tenglish: Manam rules once set chestham. Tarvata S3 automatic ga objects ni cheaper storage class ki move cheyyachu or delete cheyyachu.

Example: After 30 days → Standard-IA → After 90 days → Glacier → After 365 days → Delete.

6. Lifecycle Rule — 6 Main Actions

Action	Simple meaning	Tenglish memory
Transition current	Move live/current object to a cheaper class.	Live object ni cheaper class ki move.
Transition noncurrent	Move old versions to a cheaper class.	Old versions ni cheaper class ki move.
Expire current	Expire current object; with versioning it can add a Delete Marker.	Current object expire; versioning unte marker.
Delete noncurrent	Permanently remove old versions.	Old versions ni actual ga delete.
Delete expired markers	Clean up unnecessary expired Delete Markers.	Unnecessary markers cleanup.
Abort incomplete uploads	Clean up failed/incomplete multipart uploads.	Complete kaani uploads ni cleanup.

Super Memory: Current → Transition | Noncurrent → Transition/Delete | Expire → Delete Marker | Abort → Failed upload cleanup.

7. Static Website Hosting

English: A static website contains HTML, CSS, JavaScript and images, without backend processing or server-side logic.

Tenglish: Static website lo mainly HTML, CSS, JS, images untayi. Backend/database/server-side logic undadu.

Basic flow: Create S3 bucket → Upload index.html/error.html/CSS/JS → Enable static website hosting → Configure access policy → Use website URL.

Optional: CloudFront can be placed in front for CDN and HTTPS benefits.

8. S3 Storage Classes

Storage Class	Simple meaning	Example
Standard	Frequently accessed	Website images, active app files
Standard-IA	Rarely accessed	Backups, disaster recovery
One Zone-IA	Rare + single AZ	Secondary/re-creatable data
Intelligent-Tiering	AWS moves based on access	Unknown access pattern
Glacier Instant Retrieval	Archive + fast retrieval	Medical/archive images
Glacier Flexible Retrieval	Archive + slower retrieval	Long-term backups
Glacier Deep Archive	Very long-term archive	Legal/compliance records

Easy selection:

Frequent → Standard

Rare → Standard-IA

Single AZ + cheaper → One Zone-IA

Unknown pattern → Intelligent-Tiering

Archive + fast → Glacier Instant

Archive + low cost → Glacier Flexible

Very long-term → Deep Archive

9. Durability vs Availability

Durability: "AWS data ni lose chestunda?" S3 classes in the source notes provide 11 nines durability.

Availability: "Data ni ippudu access cheyyagalama?" Availability differs by storage class.

Memory: Durability = Data safe? | Availability = Data accessible?

10. Minimum Storage Duration

The source notes list these minimum durations:

Class	Minimum storage duration
Standard-IA	30 days
One Zone-IA	30 days
Glacier Instant Retrieval	90 days
Glacier Flexible Retrieval	90 days
Deep Archive	180 days

If an object is deleted or transitioned before a class's minimum period, the minimum-period billing rule can still apply.

11. S3 vs EBS vs EFS

Feature	S3	EBS	EFS
Type	Object	Block	File
Access	HTTP-based	Attached disk	NFS
EC2 access	Not attached	One EC2 at a time	Many EC2 instances
Example	Files/objects	EC2 disk	Shared filesystem

Memory: S3 = Object | EBS = Block | EFS = File

12. Real DevOps Uses of S3

1. **Static website + CI/CD:** Developer → GitHub → CI/CD build → S3 → CloudFront → Users.
2. **Log storage:** EC2/containers → logs → S3 → Athena/monitoring; lifecycle can move old logs to Glacier.
3. **Backup & Disaster Recovery:** DB backup/dump → S3 → another Region; versioning and lifecycle help manage backups.
4. **Terraform remote state:** S3 bucket + versioning + encryption; the source notes also mention DynamoDB for state locking.
5. **CI/CD artifacts:** Source code → Build → JAR/WAR/Docker layers → S3 → Deployment.

13. Final Interview Revision

English: Amazon S3 is an AWS object storage service used to store and retrieve data at scale. Data is stored as objects inside buckets. S3 provides high durability, versioning, lifecycle management and multiple storage classes. It is commonly used for static websites, backups, logs, Terraform state and CI/CD artifacts.

Tenglish: S3 anedi AWS object storage service. Files/data ni buckets lo objects ga store chestham. Versioning tho old versions protect cheyyachu, lifecycle policies tho old data ni cheaper classes ki move cheyyachu. S3 ni websites, backups, logs, Terraform state and CI/CD artifacts kosam use chestham.

ONE-LINE MEMORY: S3 = Scalable Object Storage + Versioning + Lifecycle + Multiple Storage Classes + DevOps Use Cases.